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TEAM COVER PAGE

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A Sector-Scalable Generative Quantum Eigensolver with Adaptive Topology and Exact Structured Circuit Synthesis

Team Quantum Pattern Recognition

GIC 2026 -- Mitsubishi Chemical Group / AIST -- Scaling GQE Using NVIDIA CUDA-Q

Executive summary

Ground-state energies support molecular and materials screening, but fixed-ansatz variational quantum eigensolvers (VQEs) trade expressivity against circuit and optimization cost, while generative quantum eigensolvers (GQEs) add repeated candidate evaluation and hardware-feasibility constraints [1-4]. We implement a molecule-parameterized workflow combining deterministic active-space tensors, a particle- and spin-projection-preserving Givens backbone, a hard-masked CUDA-QX Transformer residual, exact fixed-sector evaluation, topology screening, and exact pair-double synthesis. RHF is the same-instance classical baseline and CASCI the exact active-space referee. The optimized Givens backbone--not the residual search--reaches 0.000101 mHa for BeH2-6, 0.6492 mHa for BeH2-12, and 0.000831 mHa for partitioned LiH-40. The residual GQE evaluates 126 candidates at 6, 12, and 40 qubits but retains identity in every matched run. Exact structured synthesis reduces logical CX by 82.0-82.6% and depth by 83.8-85.2% without changing the represented state. We do not claim computational quantum advantage: the largest tested sector has D=3,136, and the 40-qubit result is a classical D=400 fixed-sector evaluation, not QPU execution.

1. Problem, state of the art, and research question

The challenge is to scale GQE toward useful active spaces while retaining accuracy, executable circuits, and a judge-reproducible audit trail. Chemistry VQE relies on a chosen ansatz and can be difficult to optimize [4]. Original Transformer-GQE automated molecular circuit generation [1]; constrained GQE subsequently bounded circuit-cutting overhead [2], and QSCI-guided GQE optimized determinant-subspace quality [8]. These advances do not jointly resolve deterministic molecular replay, symmetry enforcement, bounded cutting risk, topology choice, structured export, and finite-shot validation. Moreover, circuit cutting exchanges width for quasi-probability sampling overhead [2,3,9], while a nominal 40-qubit dense state still requires approximately 1.10 trillion amplitudes.

Our earlier Givens-exchange ansatz supplies the particle-preserving backbone [6], and our resource-continuation audit motivates treating topology and complete execution cost as explicit variables [12]. We ask whether GQE can operate at 40+ logical qubits by exploiting declared molecular symmetries while exposing--rather than hiding--the costs of search, synthesis, topology, cutting, sampling, and warm starts.

1.1 Contributions

Component	Contribution	Purpose
Molecular model	Twice-generated, checksummed active-space tensors	Deterministic judge replay
Classical comparators	Same-instance RHF and exact CASCI	Fair error attribution
Constrained GQE	Symmetry and cutting-budget masks before sampling	Score only feasible circuits
Topology / export	Balanced-root Pareto screen; exact pair-double synthesis	Expose interface cost; reduce CX/depth
Evidence	Warm/cold, QSCI, QPD, CUDA-Q, Aer, and QPU controls	Retain only supported claims

Workflow. PySCF tensors -> Givens backbone -> hard-masked Transformer residual -> exact-sector energy -> structured export and CUDA-Q/Aer/qBraid checks. The classical layer performs chemistry, policy training, optimization, and validation; the output remains an explicit quantum circuit with declared depth, sampling, and cutting costs.

2. Method

2.1 Molecular sector and deterministic evaluator

PySCF constructs an active-space electronic Hamiltonian H and the restricted-Hartree-Fock (RHF) and complete-active-space configuration-interaction (CASCI) references on the identical problem [5]. For m active spatial orbitals and separately conserved alpha and beta electron counts, the 2m-qubit state occupies D=C(m,Nalpha)C(m,Nbeta) determinants. PySCF contraction kernels evaluate E=<psi,H psi> directly in that sector. Large-width classical cost therefore follows D rather than an array over every computational-basis state. This is a low-filling strategy, not a generic cure for exponential scaling: at m=20, D=400 for (1,1), but D=34,134,779,536 for (10,10). Every frozen bundle records geometry, basis, charge, spin, frozen core, orbital order, sector, CASCI energy, and SHA-256 fingerprints; two single-thread builds had to match exactly.

2.2 Particle-preserving backbone and residual GQE

Two layers of equal-spin Givens exchanges and paired double exchanges conserve Nalpha, Nbeta, and spin projection [4,6]. Adam (learning rate 0.05), optional L-BFGS polishing, and declared adaptive depth optimize this fixed Givens-VQE backbone. A CUDA-QX Transformer then generates a six-token residual. Each token is identity, an alpha- or beta-spin exchange, or a paired double, with an orbital pair and signed angle from (+/-0.0125, +/-0.025, +/-0.05). The full policy uses embedding dimension 96, 25 iterations, five samples per iteration, and the identity control: 126 zero-shot exact-sector candidate evaluations. A generated residual is accepted only if it improves the frozen backbone.

2.3 Hard cutting and symmetry budgets

Cross-interface actions accumulate the constrained quasi-probability factor Phi [2,3]. Before sampling, the policy masks any token that would break particle number or spin projection, leave the residual vocabulary, or make the measured backbone-plus-residual cost exceed Phi_max=15. Because reconstruction variance is associated with an approximate Phi^2 multiplier, Phi is a risk indicator--not an achieved speed-up. Tests require zero mask violations.

2.4 Entanglement-informed topology

A neutral all-pair pilot supplies Schmidt information. Balanced orbital bipartitions are enumerated at small width and explored by deterministic swaps at larger width. Candidates are screened jointly by Schmidt entropy S, Hamiltonian cross weight Jcross, and Phi. A changed split must differ from the incumbent, improve at least one declared screen metric, remain on the Pareto front and within Phi_max, and be energy-noninferior within 0.1 mHa. This single-root search is motivated by our topology audit [12] and inspired by local tree-tensor-network reconnection [11]; it is not Hikihara et al.'s automatic TTN algorithm, and lower entropy is not an energy guarantee.

2.5 Structured synthesis, warm starts, and QSCI gate

The generic four-qubit pair-double unitary is replaced by an exact Pauli-network decomposition. Whole-state fidelity is checked at 6 and 12 qubits; exact primitive composition is checked at 40 qubits without allocating a dense state. Depth warm starts append identity gates; active-space and topology warm starts copy shared effective pair angles and initialize new pairs at identity. Embedded fidelity, final-rung calls, and complete cascade cost are reported separately. Quantum-selected configuration interaction (QSCI) samples symmetry-valid determinants and diagonalizes H in a capped K-determinant subspace [8]. QSCI is tested as readout and policy objective. Promotion requires matched budgets, at least 0.1 mHa improvement, a paired bootstrap 95% upper bound below zero, no material raw-state degradation, and zero mask violations.

3. Experimental design and reproducibility

3.1 Systems and roles

System	Geometry / basis	Active e / orbitals	Qubits / D	Role
H2	R=0.7474 Angstrom; STO-3G	2 / 2	4 / 4	GQE + QPU diagnostic
BeH2	linear R=1.33 Angstrom; correlation-consistent double-zeta	4 / 3,6,8	6,12,16 / 9,225,784	accuracy ladder
H2O	frozen Cartesian; correlation-consistent double-zeta	8 / 6	12 / 225	held-out portability
N2	R=1.0977 Angstrom; correlation-consistent double-zeta	10 / 8	16 / 3,136	correlation stress
LiH	R=2.20 Angstrom; augmented correlation-consistent double-zeta	2 / 20,22,24	40,44,48 / 400,484,576	low-filling scale

The procedure is molecule-parameterized: a compatible Hamiltonian rebuilds the sector, determinant, topology scores, vocabulary, and parameters. Portability means re-instantiation and reoptimization--not zero-shot transfer or guaranteed accuracy. Only LiH is demonstrated at 40-48 qubits; H2O is held out and N2 is a correlation stress test. Full coordinates and integral fingerprints are supplied.

3.2 Comparators, metrics, and controls

RHF is the non-quantum mean-field baseline; CASCI is the same-active-space exact referee. The optimized Givens circuit is the fixed-ansatz VQE baseline and identity control for residual GQE. Error is method energy minus CASCI energy, and chemical accuracy is set at 1.6 mHa. Multi-seed controls use 17, 42, and 3047 and report medians; single-seed scaling rows are labelled. Metrics include sector dimension, energy evaluations, wall time, Phi, logical CX/depth, fidelity, selected determinants, shots, uncertainty, and hardware cost. Exact-energy and QSCI policies use the same fingerprint, topology, Transformer, six-token sequence, and 126-candidate ceiling. Held-out QSCI uses K=4/16/32 and 2,000/10,000/10,000 shots at 6/12/40 qubits, each over 30 unseen trials. Feasible-random and stratified-greedy controls receive the same 126-evaluation ceiling; identity is a fixed zero-evaluation reference.

3.3 Platform roles

Stage	Role	Implementation
Chemistry	Frozen tensors; RHF/CASCI referee	PySCF 2.14.0 [5]
Backbone	Exact-sector optimization	NumPy / PyTorch
Generative	Hard-masked residual search	CUDA-QX Solvers 0.6.0
Checks	State convention / sampled energy	CUDA-Q 0.14.2 qpp-cpu / Aer
Delivery	Credential-free Run-All; archived QPU payload	qBraid Lab [10]

3.4 Judge reproduction

The archive supplies frozen tensors, a complete 126-distribution Python 3.12/Linux lock, 48 core regressions, advanced tests, declared tolerances, and a qBraid Run-All notebook. The default path uses CPU simulation, imports no provider, and cannot spend credits. It records platform, lock hash, actual workflow wall time, and qpu_contacted=false. CUDA-Q full-register checks are limited to safe widths; 40-48-qubit costs use the fixed-sector callback and never allocate dense statevectors. First setup requires Internet access for packages but no private data, GPU, provider credential, or QPU allocation.

4. Results and controlled decisions

Table 1. Same-instance classical and optimized Givens-backbone baselines. Error is relative to CASCI; times cover Givens-VQE optimization only and are platform-dependent.

Configuration	D	Seeds	RHF error (mHa)	Givens error (mHa)	Time (s)
BeH2 6q, partitioned	9	3	1.0215	0.000101	0.28
BeH2 12q, partitioned	225	3	3.9727	0.6492	1.04
BeH2 16q, all-pair	784	1	8.3516	0.1867	8.49
H2O 12q, held out	225	3	6.5314	3.1289	1.11
N2 16q, adaptive depth	3,136	1	80.2523	2.8080	36.51
LiH 40q, partitioned	400	3	16.3861	0.000831	15.71
LiH 40q, all-pair	400	1	16.3861	6.768e-8	33.72
LiH 44q / 48q, all-pair	484 / 576	1 / 1	18.9392 / 19.1976	6.806e-8 / 1.580e-5	39.73 / 53.90

H2O and N2 exceed the 1.6 mHa target, showing that sector compression does not guarantee ansatz accuracy. LiH tractability follows its two-active-electron sector; it does not imply generic 48-qubit simulation. Each variational energy is exact within floating-point arithmetic for the declared sector, but optimization is not guaranteed to reach CASCI. The 44q and 48q rows are single-seed backbone scaling controls--not residual-GQE searches.

Table 2. Matched residual GQE: 126 zero-shot exact-sector candidate evaluations per Transformer arm; all errors are mHa.

System	Frozen backbone	Sampled nonidentity	Guarded result	Selection
BeH2 6q	3.448e-8	0.001049	3.448e-8	identity
BeH2 12q	0.651671	0.653761	0.651671	identity
LiH 40q	0.000685	0.001263	0.000685	identity

No six-token residual beat its frozen backbone; the identity guard prevented degradation. Feasible-random and stratified-greedy controls used the same 126-evaluation ceiling and also retained identity. This establishes executable constrained search, not residual-search advantage. Separately, four-qubit H2 Pauli-pool GQE selected a nonidentity ten-operator circuit and reduced ideal error from RHF 20.8501 to 0.1337 mHa (99.36%).

Table 3. Logical all-to-all export of identical frozen matched-topology Givens states: generic versus exact structured synthesis.

Width	Generic CX/depth	Structured CX/depth	CX/depth reduction	Equivalence
6q	374 / 1,278	65 / 192	82.62% / 84.98%	whole-state F approximately 1
12q	1,325 / 2,316	235 / 343	82.26% / 85.19%	whole-state F approximately 1
40q	17,190 / 9,483	3,091 / 1,537	82.02% / 83.79%	exact primitives

Adaptive Pareto splits were accepted at BeH2-6/12 and rejected at held-out H2O-12 and LiH-40; lower entropy sometimes increased cross-Hamiltonian weight, so no universal topology benefit is claimed. Warm starts achieved unit embedding fidelity, but complete cascades cost 2.172x a direct final-rung solve. QSCI retained identity at 6/12q; at LiH-40 it worsened raw error by 1.92036 mHa, while the paired held-out median change -0.01181 mHa had 95% interval [-0.14662,0.12754]. The interval crosses zero, so exact energy remains the objective.

5. Cross-platform evidence, quantum benefit, and limitations

Table 4. Cross-platform reproduction evidence and declared execution budgets.

Check	Mode / budget	Result	Boundary
H2 4q GQE	CUDA-QX qpp-cpu; 25x5 samples	0.1337 mHa; 58 est. 2q gates	separate Pauli-pool branch
BeH2 6q sampled energy	Aer; 9x20,000 shots	0.8575 mHa; SE=2.242 mHa	one finite-shot draw
BeH2 6q exact QPD	4+2; 10,000 branches	abs(dE) <= 1e-10 Ha	algebraic, not sampled
BeH2 6q / 12q	CUDA-Q qpp-cpu	abs(dE) <= 1e-10 Ha	state-convention check
H2 4q QPU	Forte-1 via qBraid; 350 shots	one Pauli expectation	not an energy

5.1 Physical four-qubit diagnostic

The frozen H2 Pauli-pool Transformer-GQE circuit measured W=X0xX1xY2xY3 on four logical qubits. Ideal simulation gives <W>=-0.210899; 350 requested and returned IonQ Forte-1 shots give -0.142857 with exact 95% interval [-0.247808,-0.035469] and shot-noise standard error 0.052904. The RHF determinant has <W>=0, so the interval supports nonzero phase-sensitive coherence relative to RHF and contains the ideal value. The submitted measurement circuit had 58 source-level CX gates and logical depth 108; the task used 2,830 qBraid credits and provider-recorded duration 277.264 s. Provider-native transformed depth was not retained. This is one predeclared observable--not an energy, fidelity, entanglement certificate, chemical-accuracy result, or quantum-advantage demonstration. The QPU did not measure the ideal H2 energy reported above.

5.2 Demonstrated quantum benefit and advantage boundary

CASCI exactly solves every tested active-space sector, whose largest dimension is D=3,136; therefore, we do not claim computational quantum advantage. The demonstrated quantum-specific benefit is implementation-level: exact structured synthesis reduces logical CX by 82.0-82.6% and depth by 83.8-85.2% relative to generic export of the same states, while hard masks reject symmetry-breaking or Phi>15 actions before evaluation. These are circuit-resource and feasibility benefits, not evidence that the quantum workflow outperforms the best classical solver. The counts are Qiskit optimization-level-3 logical all-to-all diagnostics (seed 3047), not target-native forecasts.

5.3 Cutting boundary, limitations, and conclusion

The LiH 20+20 projection defines two 20-spin-qubit fragments with theoretical Phi=2.4851 and Phi^2=6.1756, but cross-pair-double QPD, sampled reconstruction, joint fragment QSCI, and paid execution were not performed. Adding two independent fragment energies would omit cross-fragment correlation; exact QPD was executed only for the 6q 4+2 control. More broadly, 40-48q results are low-filling classical sector evaluations; H2O and N2 miss chemical accuracy; residual GQE does not beat the backbone; topology is not uniformly beneficial; complete warm-start cascades cost more; QSCI fails promotion; Aer noise is not target-calibrated; and logical counts are not hardware-native predictions.

For Mitsubishi Chemical Group and AIST, the result is an auditable eigensolver component for localized active spaces in molecular and materials-relevant fragments--not a complete screening pipeline. Deterministic inputs, same-instance classical baselines, feasibility masks, exact structured export, and independent simulator/QPU checks provide a bounded route toward an enhanced six-qubit hardware pilot. That pilot should proceed only after live backend-native compilation, current-calibration noise analysis, a predeclared readout, power analysis, and a current cost check [10].

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All numerical claims are regenerated from the supplied frozen inputs and checked against Source_Code/expected_metrics.json; full precision, hashes, and tolerances remain machine-readable.